# PRODUCT NAME

## Advanced Under Arm Whitening Cream Formula

## Description

A professional cream formulation designed for underarm whitening with enhanced skin lightening efficacy. This product is gentle on the skin, hypoallergenic, moisturizing, and allows for fast absorption. It is paraben-free and offers a non-greasy feel, ideal for daily use.

## Technical Specifications

| | |
|---|---|
| **pH Level:** | 7 |
| **Viscosity:** | High |
| **Shelf Life:** | 24 months |
| **Batch Size:** | 10-100 kg |
| **Processing Time:** | 2.5 - 3 hours |
| **Temperature:** | 75°C for emulsification, 40°C for additive phase, room temperature for final phase |
| **Storage Conditions:** | Store in a cool, dry place away from direct sunlight. |
| **Equipment:** | Stainless steel mixing vessels, high-shear mixer, heating system, cooling system |
| **Certification:** | GMP, ISO 22716 |

## Formulation Table

| Sr.No | Ingredient | INCI Name | % | Function |
|---|---|---|---|---|
| 1 | **Distilled Water** | Aqua | 57.5% | Base solvent |
| 2 | **Glycerin** | Glycerin | 8.0% | Moisturizer, humectant |
| 3 | **Niacinamide** | Niacinamide | 5.0% | Skin lightening agent, anti-inflammatory |
| 4 | **Cetearyl Alcohol** | Cetearyl Alcohol | 7.0% | Thickening agent, stabilizer |
| 5 | **Ceteareth-20** | Ceteareth-20 | 2.0% | Emulsifier |
| 6 | **Shea Butter** | Butyrospermum Parkii Butter | 6.5% | Moisturizer, skin conditioner |
| 7 | **Sweet Almond Oil** | Prunus Amygdalus Dulcis Oil | 5.0% | Emollient |
| 8 | **Titanium Dioxide** | Titanium Dioxide | 3.0% | Whitening agent, opacity enhancer |
| 9 | **Phenoxyethanol and Ethylhexylglycerin** | Phenoxyethanol, Ethylhexylglycerin | 1.0% | Preservative |
| 10 | **Fragrance** | Parfum | 1.0% | Aesthetic enhancer |
| 11 | **Xanthan Gum** | Xanthan Gum | 1.0% | Stabilizer, viscosity enhancer |
| 12 | **Disodium EDTA** | Disodium EDTA | 0.5% | Chelating agent |
| **Total** | | | **100%** | |

## Manufacturing Process

### Phase 1: Aqueous Phase

1. Heat distilled water to 75°C in a stainless steel mixing vessel.
2. Add Glycerin and mix until completely dissolved.
3. Hydrate Xanthan Gum into the mixture until a homogenous solution is obtained.

**Phase 2: Oil Phase**

1. In a separate vessel, heat Shea Butter, Sweet Almond Oil, Cetearyl Alcohol, and Ceteareth-20 to 75°C.

2. Mix until all solid components are completely melted and the phase is uniform.

**Phase 3: Emulsification Phase**

1. Slowly add the oil phase to the aqueous phase with continuous stirring using a high-shear mixer.

2. Maintain temperature at 70°C and mix for 15 minutes to ensure a stable emulsion is formed.

**Phase 4: Cooling and Additive Phase**

1. After emulsification, start cooling the mixture to 40°C while stirring gently.

2. Add Niacinamide, Titanium Dioxide, Phenoxyethanol and Ethylhexylglycerin, and Disodium EDTA.

3. Add fragrance and mix until homogenous.

**Phase 5: Final Adjustments**

1. Continue mixing until the temperature reaches room temperature.

2. Check and adjust the pH to ensure the final product has a pH of 7.

## Required Equipment

Stainless steel mixing vessels, high-shear mixer, heating system, cooling system

## Safety Precautions

**Handling:** Wear appropriate PPE including gloves, safety glasses, lab coat.

**PPE Requirements:** Chemical-resistant gloves, safety goggles, protective clothing.

**Storage:** Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

**Storage Conditions:** Store in a cool, dry place away from direct sunlight.

## Packaging Notes

**Packaging:** Use chemically compatible containers (HDPE, glass, or PET).

**Labeling:** Include product name, ingredients, usage instructions, and safety warnings.

**Certification:** GMP, ISO 22716

## Scaling Note

**Lab Scale:** This formulation is designed for laboratory testing and development.

**Pilot Scale:** For pilot production, scale proportionally and verify all parameters.

**Production Scale:** Consider equipment limitations, mixing efficiency, and process validation.

**Batch Size:** Current formulation is optimized for 10-100 kg.

**Scaling Factor:** Maintain ingredient ratios while adjusting processing parameters as needed.

## Product Usage Instructions & Application Guidelines

Apply a small amount to clean, dry underarm area twice daily. Massage gently until fully absorbed. Use consistently for best results.